

Snake-inspired soft robot that can burrow through sand

We have seen robots take to the air, dive beneath the waves, and perform all sorts of manoeuvres on land. Taking cues from plants and animals that have evolved to navigate subterranean spaces, researchers at UC Santa Barbara and Georgia Institute of Technology have now developed a fast, controllable soft robot that can burrow through the sand. The technology not only enables new applications for fast, precise, and minimally invasive movement underground but also lays mechanical foundations for new types of robots. A small, exploratory, soft robot such as this has a variety of applications where shallow burrowing through dry granular media is needed, such as soil sampling, underground installation of utilities, and erosion control.



Researchers have developed a fast, steerable, burrowing soft robot (Credit: www.news.ucsb.edu)

Cement based rechargeable batteries



Rechargeable cement-based batteries utilised as functional concrete (Credit: www.chalmers.se/en)

The ever-growing need for sustainable building materials poses great challenges for researchers. Now thanks to unique research from Chalmers University of Technology, Sweden, the technology for rechargeable cement based battery has been conceptualised. It involves a cement based mixture having small amounts of short carbon fibres for increased conductivity and flexural toughness. Embedded within the mixture is a metal-coated carbon fibre mesh—iron for the anode and nickel

for the cathode. The rechargeable cement based battery has an average energy density of 7 watthours per square metre (or 0.8 watthours per litre). The energy density is still low in comparison to commercial batteries, but this limitation could be overcome by huge volume utilisation while constructing buildings. It could also be coupled with solar cell panels to provide electricity for monitoring systems in highways or bridges.